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Flexible support for aircraft motor bearing

Abstract

An assembly is provided for an aircraft motor. This assembly includes a bearing, a stationary structure and a flexible support. The bearing extends circumferentially around an axis. The stationary structure circumscribes the bearing. The flexible support is arranged radially between and radially engages the bearing and the stationary structure. The flexible support includes a first ring, a second ring, a plurality of bridges and a plurality of first ribs. The bridges are arranged circumferentially about the axis. Each of the bridges extends radially between and is connected to the first ring and the second ring. The first ribs are arranged circumferentially about the axis and interspersed with the bridges. Each of the first ribs projects radially out from the first ring towards the second ring. Each of the first ribs is connected to the first ring and is disengaged from the second ring.

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Background/Summary

TECHNICAL FIELD

(1) This disclosure relates generally to an aircraft motor and, more particularly, to a bearing support member for use within the aircraft motor.

BACKGROUND INFORMATION

(2) An aircraft motor such as a gas turbine motor may include a flexible bearing support to accommodate slight radial shifts between a rotating structure and a stationary structure of the aircraft motor. Various types and configurations of flexible bearing supports are known in the art. While these known flexible bearing supports have various benefits, there is still room in the art for improvement.

SUMMARY

(3) According to an aspect of the present disclosure, an assembly is provided for an aircraft motor. This assembly includes a bearing, a stationary structure and a flexible support. The bearing extends circumferentially around an axis. The stationary structure circumscribes the bearing. The flexible support is arranged radially between and radially engages the bearing and the stationary structure. The flexible support includes a first ring, a second ring, a plurality of bridges and a plurality of first ribs. The bridges are arranged circumferentially about the axis. Each of the bridges extends radially between and is connected to the first ring and the second ring. The first ribs are arranged circumferentially about the axis and interspersed with the bridges. Each of the first ribs projects radially out from the first ring towards the second ring. Each of the first ribs is connected to the first ring and is disengaged from the second ring.

(4) According to another aspect of the present disclosure, another assembly is provided for an aircraft motor. This assembly includes a bearing, a stationary structure and a flexible support. The bearing extends circumferentially around an axis. The stationary structure circumscribes the bearing. The flexible support is arranged radially between the bearing and the stationary structure. The flexible support includes a first ring, a second ring, a plurality of bridges and a plurality of first ribs. One of the first ring and the second ring radially engages the bearing. The other one of the first ring and the second ring radially engages the stationary structure. The bridges are arranged circumferentially about the axis. Each of the bridges extend radially between and is connected to the first ring and the second ring. The first ribs are arranged circumferentially about the axis and are interspersed with the bridges. Each of the first ribs projects radially out from the first ring towards the second ring. Each of the first ribs is connected to the first ring and is disengaged from the second ring.

(5) According to still another aspect of the present disclosure, an apparatus is provided for an aircraft motor. This apparatus includes a flexible support for a bearing in the aircraft motor. The flexible support includes a plurality of members extending axially along an axis from a first side of the flexible support to a second side of the flexible support. The members include a first ring, a second ring, a plurality of first bridges and a plurality of first ribs. The first ring extends circumferentially around the axis. The second ring extends circumferentially around the axis. The first bridges are arranged circumferentially about the axis. Each of the first bridges projects radially from the first ring to the second ring. The first ribs are arranged circumferentially about the axis and interspersed with the first bridges. Each of the first ribs projects radially out from the first ring, in a direction towards the second ring, to a respective unsupported first rib distal end. A plurality of first ports extend axially through the flexible support from the first side of the flexible support to the second side of the flexible support. Each of the first ports extends radially within the flexible support from the first ring to the second ring. Each of the first ports extends circumferentially within the flexible support between a respective one of the first bridges and a respective one of the first ribs.

(6) The members may also include a third ring, a plurality of second bridges and a plurality of second ribs. The third ring may extend circumferentially around the axis with the second ring radially between the first ring and the third ring. The second bridges may be arranged circumferentially about the axis. Each of the second bridges may project radially from the second ring to the third ring. The second ribs may be arranged circumferentially about the axis and interspersed with the plurality of second bridges. Each of the second ribs may project radially out

from the second ring, in a direction towards the third ring, to a respective unsupported second rib distal end.
(7) A plurality of second ports may extend axially through the flexible support from the first side of the flexible support to the second side of the flexible support. Each of the second ports may extend radially within the flexible support from the second ring to the third ring. Each of the second ports may extend circumferentially within the flexible support between a respective one of the second bridges and a respective one of the second ribs.
(8) The first ring may be an inner ring. The second ring may be an outer ring which circumscribes the inner ring.
(9) The second ring may be an inner ring. The first ring may be an outer ring which circumscribes the inner ring.
(10) The flexible support may also include a plurality of second ribs. Each of the second ribs may be connected to and project radially out from the second ring. Each of the second ribs may be circumferentially aligned with a respective one of the first ribs. Each of the second ribs may be separated from the respective one of the first ribs by a respective radial gap.
(11) A first of the first ribs may be circumferentially aligned with a first of the second ribs. A radial height of the first of the first ribs may be equal to a radial height of the first of the second ribs.
(12) A first of the first ribs may be circumferentially aligned with a first of the second ribs. A radial height of the respective radial gap radially between and formed by the first of the first ribs and the first of the second ribs may be less than a radial height of the first of the first ribs and/or a radial height of the first of the second ribs.
(13) A first of the first ribs may be circumferentially aligned with a first of the second ribs. A radial height of the respective radial gap radially between and formed by the first of the first ribs and the first of the second ribs may be less than a lateral thickness of the first of the first ribs and/or a lateral thickness of the first of the second ribs.
(14) Each of the first ribs may be separated from the second ring by a respective radial gap.
(15) A radial height of the respective radial gap radially between and formed by a first of the first ribs and the second ring may be less than: a radial height of the first of the first ribs; and/or a lateral thickness of the first of the first ribs.
(16) A radial height of a first of the bridges may be: equal to or less than a lateral width of the first of the bridges; and/or greater than an axial width of the first of the bridges.
(17) A radial height of a first of the bridges may be greater than a radial thickness of the first ring and/or a radial thickness of the second ring.
(18) The flexible support may extend axially between a support first side and a support second side. The first ring, the second ring, each of the bridges and each of the first ribs may project axially to the support first side. In addition or alternatively, the first ring, the second ring, each of the bridges and each of the first ribs may project axially to the support second side.
(19) The bearing may be configured as or otherwise include a rolling element bearing with an outer race. One of the first ring or the second ring may circumscribe and radially engage the outer race.
(20) One of the first ring or the second ring may circumscribe and may radially contact the bearing.
(21) The stationary structure may circumscribe and may radially contact one of the first ring or the second ring.
(22) The flexible support may be attached to the stationary structure through an interference fit at a radial interface between the stationary structure and one of the first ring or the second ring.
(23) The flexible support may be attached to the stationary structure through a bonded connection between the stationary structure and one of the first ring or the second ring.
(24) The flexible support may be attached to the stationary structure with one or more fasteners.
(25) The bridges may be a plurality of first bridges, and the flexible support may also include a third ring, a plurality of second bridges and a plurality of second ribs. The second ring may be radially between the first ring and the third ring. The second bridges may be arranged

circumferentially about the axis. Each of the second bridges may extend radially between and may be connected to the second ring and the third ring. The second bridges may be circumferentially offset from the first bridges. The second ribs may be arranged circumferentially about the axis and interspersed with the second bridges. Each of the second ribs may project radially out from the second ring towards the third ring. Each of the second ribs may be connected to the second ring and may be disengaged from the third ring.
(26) The present disclosure may include any one or more of the individual features disclosed above and/or below alone or in any combination thereof.
(27) The foregoing features and the operation of the invention will become more apparent in light of the following description and the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a sectional illustration of a portion of an assembly for an aircraft motor.
(2) FIG. **2** is a sectional illustration of a portion of a flexible support at a first circumferential position.
(3) FIG. **3** is a sectional illustration of a portion of the flexible support at a second circumferential position.
(4) FIG. **4** is a side view illustration of a portion of the flexible support.
(5) FIG. **5** is a sectional illustration of a portion of the assembly with another flexible support mounting arrangement.
(6) FIGS. **6** and **7** are partial perspective illustrations of the flexible support with other arrangements of support ribs.
(7) FIGS. **8** and **9** are partial perspective illustrations of the flexible support with multiple layers of support bridges and support ribs.
(8) FIG. **10** is a sectional illustration of a portion of another assembly for the aircraft motor.
DETAILED DESCRIPTION
(9) FIG. **1** illustrates an assembly **20** for a motor of an aircraft. The aircraft may be an airplane, a helicopter, a drone (e.g., an unmanned aerial vehicle (UAV)) or any other manned or unmanned aerial vehicle or system. The aircraft motor may be configured as, or otherwise included as part of, a propulsion system for the aircraft. The aircraft motor may also or alternatively be configured as, or otherwise included as part of, an electrical power system for the aircraft. The motor assembly **20** of FIG. **1** includes a stationary structure **22** of the aircraft motor and a rotating structure **24** of the aircraft motor. This motor assembly **20** also includes a mounting assembly **26** for rotatably mounting the rotating structure **24** to the stationary structure **22**. The mounting assembly **26** of FIG. **1** includes a bearing **28** and a flexible support **30** for the bearing **28**; e.g., a flexible bearing support.
(10) The stationary structure **22** may be configured as, or otherwise included as part of, a casing structure of the aircraft motor. The stationary structure **22** of FIG. **1**, for example, includes a motor case **32**, a stationary outer mounting land **34** and a land support **36** (e.g., an annular frame, an array of struts, etc.) extending radially between and structurally tying the outer mounting land **34** to the motor case **32**, where the motor case **32** and the land support **36** are schematically shown. The outer mounting land **34** extends axially along an axis **38**. Briefly, this axis **38** may be a centerline axis of the aircraft motor and/or one or more of its members **22**, **24**, **26**, **28** and/or **30**. The axis **38** may also or alternatively be a rotational axis of the rotating structure **24**. The outer mounting land **34** projects radially inward (towards the axis **38**) to a radial inner side **40** of the outer mounting land **34**. The outer mounting land **34** extends circumferentially about (e.g., completely around) the axis **38**, which provides the outer mounting land **34** with, for example, a full-hoop (e.g., tubular or annular) geometry.

(11) The outer mounting land **34** of FIG. **1** includes an interior land surface **42** disposed at the land inner side **40**. This interior land surface **42** may have a regular cylindrical geometry. The land inner side **40** and its interior land surface **42** partially or completely form a radial outer peripheral boundary of an internal bore **44** in the stationary structure **22** and its outer mounting land **34**. This land bore **44** extends axially along the axis **38** in (e.g., into, within or through) the stationary structure **22** and its outer mounting land **34**.
(12) The rotating structure **24** may be configured as, other otherwise included as part of, a spool or other rotating assembly of the aircraft motor. The rotating structure **24** of FIG. **1**, for example, includes an inner mounting land **46** extending axially in the land bore **44**. This inner mounting land **46** may be part (e.g., an axial section) of a motor shaft. The inner mounting land **46** may alternatively be configured as, or otherwise included as part of, another component mounted onto or formed integral with the motor shaft; e.g., a shaft sleeve, a rotor disk, etc. The inner mounting land **46** extends axially along the axis **38**. The inner mounting land **46** projects radially outward (away from the axis **38**) to a radial outer side **48** of the inner mounting land **46**. The inner mounting land **46** extends circumferentially about (e.g., completely around) the axis **38**, which provides the inner mounting land **46** with, for example, a full-hoop (e.g., tubular or annular) geometry.
(13) The inner mounting land **46** of FIG. **1** includes an exterior land surface **50** disposed at the land outer side **48**. The exterior land surface **50** may have a regular cylindrical geometry.
(14) The bearing **28** may be configured as a rolling element bearing. The bearing **28** of FIG. **1**, for example, includes a bearing inner race **52**, a bearing outer race **54** and a plurality of bearing rolling elements **56** arranged circumferentially about the axis **38** in an array; e.g., a circular array. The inner race **52** extends axially along and circumferentially about (e.g., circumscribes) the inner mounting land **46** and the axis **38**. This inner race **52** is mounted onto (e.g., fixed to) and is rotatable with the rotating structure **24** and its inner mounting land **46**. The outer race **54** is spaced radially outboard from the inner race **52**. The outer race **54** extends axially along and circumferentially about (e.g., circumscribes) the inner race **52**, the array of rolling elements **56** and the axis **38**. The array of rolling elements **56** are arranged radially between the inner race **52** and the outer race **54**, where each of the rolling elements **56** is operable to radially engage and roll against the inner race **52** and the outer race **54**. The present disclosure, however, is not limited to such an exemplary rolling element bearing configuration. Moreover, while the bearing **28** is described above as the rolling element bearing, it is contemplated the bearing **28** may alternatively be configured as a journal bearing or any other type of bearing suitable for use in the aircraft motor.
(15) Referring to FIGS. **2** and **3**, the flexible support **30** extends axially along the axis **38** between and to a first side **58** of the flexible support **30** and a second side **60** of the flexible support **30**. The flexible support **30** extends radially from a radial inner side **62** of the flexible support **30** to a radial outer side **64** of the flexible support **30**. The flexible support **30** of FIG. **4** extends circumferentially about (e.g., completely around) the axis **38**, which provides the flexible support **30** with, for example, a full-hoop (e.g., annular) geometry. The flexible support **30** of FIG. **4** includes a radial inner ring **66**, a radial outer ring **68**, a plurality of bridges **70**, a plurality of inner ribs **72** and a plurality of outer ribs **74**.
(16) The support inner ring **66** is disposed at the support inner side **62**. The support inner ring **66** of FIG. **4**, for example, extends radially between and to a radial outer side **76** of the support inner ring **66** and the support inner side **62**. Referring to FIGS. **2** and **3**, the support inner ring **66** extends axially along the axis **38** between and to opposing axial sides **78** and **80** of the support inner ring **66**. The inner ring first side **78** of FIGS. **2** and **3** is aligned with the support first side **58**. The inner ring second side **80** of FIGS. **2** and **3** is aligned with the support second side **60**. The support inner ring **66** of FIG. **4** extends circumferentially about (e.g., completely around) the axis **38**, which provides the support inner ring **66** with, for example, a full-hoop (e.g., tubular) geometry.
(17) The support inner ring **66** of FIGS. **2** and **3** has an axial width **82** and a radial thickness **84**. The inner ring width **82** is measured axially along the axis **38** from the inner ring first side **78** to the

inner ring second side **80**. This inner ring width **82** may be uniform (constant) as the support inner ring **66** extends circumferentially about the axis **38**. The inner ring thickness **84** is measured radially from the support inner side **62** to the inner ring outer side **76**. This inner ring thickness **84** may be uniform as the support inner ring **66** extends circumferentially about the axis **38**.

(18) The flexible support **30** and its support inner ring **66** of FIGS. **2** and **3** include an interior support surface **86** disposed at the support inner side **62**. The interior support surface **86** may have a regular cylindrical geometry which extends axially from (or near) the inner ring first side **78** to (or near) the inner ring second side **80**.

(19) The support outer ring **68** is disposed at the support outer side **64**. The support outer ring **68** of FIGS. **2** and **3**, for example, extends radially between and to a radial inner side **88** of the support outer ring **68** and the support outer side **64**. The support outer ring **68** extends axially along the axis **38** between and to opposing axial sides **90** and **92** of the support outer ring **68**. The outer ring first side **90** of FIGS. **2** and **3** is aligned with the support first side **58** and/or the inner ring first side **78**. The outer ring second side **92** of FIGS. **2** and **3** is aligned with the support second side **60** and/or the inner ring second side **80**. The support outer ring **68** of FIG. **4** extends circumferentially about (e.g., completely around) the axis **38**, which provides the support outer ring **68** with, for example, a full-hoop (e.g., tubular) geometry.

(20) The support outer ring **68** of FIGS. **2** and **3** has an axial width **94** and a radial thickness **96**. The outer ring width **94** is measured axially along the axis **38** from the outer ring first side **90** to the outer ring second side **92**. This outer ring width **94** may be uniform (constant) as the support outer ring **68** extends circumferentially about the axis **38**. The outer ring width **94** of FIGS. **2** and **3** is equal to the inner ring width **82**. The outer ring thickness **96** is measured radially from the support outer side **64** to the outer ring inner side **88**. This outer ring thickness **96** may be uniform as the support outer ring **68** extends circumferentially about the axis **38**. The outer ring thickness **96** of FIGS. **2** and **3** is equal to the inner ring thickness **84**. The present disclosure, however, is not limited to such exemplary dimensional relationships. It is contemplated, for example, the outer ring width **94** may alternatively be sized differently (e.g., greater) than the inner ring width **82** and/or the outer ring thickness **96** may be sized differently (e.g., greater) than the inner ring thickness **84** in other embodiments.

(21) The flexible support **30** and its support outer ring **68** of FIGS. **2** and **3** include an exterior support surface **98** disposed at the support outer side **64**. The exterior support surface **98** may have a regular cylindrical geometry which extends axially from (or near) the outer ring first side **90** to (or near) the outer ring second side **92**.

(22) Referring to FIG. **4**, the support bridges **70** are arranged (e.g., and equispaced) circumferentially about the axis **38** in an array; e.g., a circular array. This array of support bridges **70** is disposed radially between the support inner ring **66** and the support outer ring **68**. Each support bridge **70** of FIG. **4**, for example, is connected to (e.g., formed integral with) the support inner ring **66** and the support outer ring **68**. Each support bridge **70** projects radially out from the support inner ring **66** at its inner ring outer side **76** to the support outer ring **68** at its outer ring inner side **88**. Referring to FIG. **3**, each support bridge **70** extends axially along the axis **38** between and to opposing axial sides **100** and **102** of the respective support bridge **70**. The bridge first side **100** of FIG. **3** is axially aligned with the support first side **58**, the inner ring first side **78** and/or the outer ring first side **90**. The bridge second side **102** of FIG. **3** is axially aligned with the support second side **60**, the inner ring second side **80** and/or the outer ring second side **92**. Each support bridge **70** of FIG. **4** extends laterally (e.g., circumferentially about the axis **38** or tangentially) between opposing lateral sides **104** and **106** of the respective support bridge **70**.

(23) Each support bridge **70** of FIG. **3** has an axial width **108** and a radial height **110**. The bridge width **108** is measured axially along the axis **38** from the bridge first side **100** to the bridge second side **102**. The bridge width **108** of FIG. **3** is equal to the inner ring width **82** and/or the outer ring width **94**. The bridge height **110** is measured radially from the support inner ring **66** at its inner ring

outer side **76** to the support outer ring **68** at its outer ring inner side **88**. The bridge height **110** of FIG. **3** is greater than the inner ring thickness **84** and/or the outer ring thickness **96**. The bridge height **110** of FIG. **3** is less than the bridge width **108**. Referring to FIG. **4**, each support bridge **70** has a lateral thickness **112** measured laterally between its opposing lateral sides **104** and **106**. The bridge thickness **112** of FIG. **4** is greater than the inner ring thickness **84**, the outer ring thickness **96** and/or the bridge height **110**. The bridge thickness **112** of FIG. **4** is less than the bridge width **108** of FIG. **3**. The present disclosure, however, is not limited to such exemplary dimensional relationships. It is contemplated, for example, the bridge height **110** of FIG. **4** may alternatively be sized equal to or greater than the bridge thickness **112** in other embodiments.

(24) The support inner ribs **72** are arranged (e.g., and equispaced) circumferentially about the axis **38** in an array; e.g., a circular array. This array of support inner ribs **72** is interspersed with the array of support bridges **70**. Each of the support inner ribs **72** of FIG. **4**, for example, is arranged circumferentially between a respective circumferentially neighboring (e.g., adjacent) pair of the support bridges **70**. Similarly, each of the support bridges **70** of FIG. **4** is arranged circumferentially between a respective circumferentially neighboring pair of the support inner ribs **72**.

(25) The array of support inner ribs **72** is disposed radially between the support inner ring **66** and the support outer ring **68**. Each support inner rib **72** of FIG. **4** is connected to (e.g., formed integral with) the support inner ring **66** and is disengaged (e.g., disconnected) from the support outer ring **68**. Each support inner rib **72** of FIG. **2**, for example, projects radially out from the support inner ring **66** at its inner ring outer side **76** to a radial outer distal end **114** of the respective support inner rib **72**. Referring to FIG. **2**, each support inner rib **72** extends axially along the axis **38** between and to opposing axial sides **116** and **118** of the respective support inner rib **72**. The inner rib first side **116** of FIG. **2** is axially aligned with the support first side **58**, the inner ring first side **78** and/or the outer ring first side **90**. The inner rib second side **118** of FIG. **2** is axially aligned with the support second side **60**, the inner ring second side **80** and/or the outer ring second side **92**. Each support inner rib **72** of FIG. **4** extends laterally between opposing lateral sides **120** and **122** of the respective support inner rib **72**.

(26) Each support inner rib **72** of FIG. **2** has an axial width **124** and a radial height **126**. The inner rib width **124** is measured axially along the axis **38** from the inner rib first side **116** to the inner rib second side **118**. The inner rib width **124** of FIG. **2** is equal to the inner ring width **82** and/or the outer ring width **94**. The inner rib height **126** is measured radially from the support inner ring **66** at its inner ring outer side **76** to the respective inner rib distal end **114**. The inner rib height **126** may be equal to (or different than) the inner ring thickness **84** and/or the outer ring thickness **96**. The inner rib height **126** of FIG. **2** is less than the inner rib width **124** and the bridge height **110** of FIG. **3**. Referring to FIG. **4**, each support inner rib **72** has a lateral thickness **128** measured laterally between its opposing lateral sides **120** and **122**. The inner rib thickness **128** of FIG. **4** is greater than the inner ring thickness **84** and/or the outer ring thickness **96**. The inner rib thickness **128** may be equal to or different than the bridge thickness **112**. The inner rib thickness **128** of FIG. **4** is less than the inner rib width **124** of FIG. **2**. The present disclosure, however, is not limited to such exemplary dimensional relationships.

(27) The support outer ribs **74** are arranged (e.g., and equispaced) circumferentially about the axis **38** in an array; e.g., a circular array. This array of support outer ribs **74** is interspersed with the array of support bridges **70**. Each of the support outer ribs **74** of FIG. **4**, for example, is arranged circumferentially between a respective circumferentially neighboring pair of the support bridges **70**. Similarly, each of the support bridges **70** of FIG. **4** is arranged circumferentially between a respective circumferentially neighboring pair of the support outer ribs **74**. Each of the support outer ribs **74** of FIG. **4** may also be circumferentially aligned with a respective one of the support inner ribs **72**. Similarly, each of the support inner ribs **72** of FIG. **4** may be circumferentially aligned with a respective one of the support outer ribs **74**.

(28) The array of support outer ribs **74** is disposed radially between the support inner ring **66** and

the support outer ring **68**. Each support outer rib **74** of FIG. **4** is connected to (e.g., formed integral with) the support outer ring **68** and is disengaged (e.g., disconnected) from the support inner ring **66** as well as the respective circumferentially aligned support inner rib **72**. Each support outer rib **74** of FIG. **2**, for example, projects radially out from the support outer ring **68** at its outer ring inner side **88** to a radial inner distal end **130** of the respective support outer rib **74**. This outer rib distal end **130** is radially spaced/separated from the opposite and circumferentially aligned inner rib distal end **114** by a radial gap **132**; an empty air gap. Each support outer rib **74** extends axially along the axis **38** between and to opposing axial sides **134** and **136** of the respective support outer rib **74**. The outer rib first side **134** of FIG. **2** is axially aligned with the support first side **58**, the inner ring first side **78** and/or the outer ring first side **90**. The outer rib second side **136** of FIG. **2** is axially aligned with the support second side **60**, the inner ring second side **80** and/or the outer ring second side **92**. Each support outer rib **74** of FIG. **4** extends laterally between opposing lateral sides **138** and **140** of the respective support outer rib **74**.
(29) Each support outer rib **74** of FIG. **2** has an axial width **142** and a radial height **144**. The outer rib width **142** is measured axially along the axis **38** from the outer rib first side **134** to the outer rib second side **136**. The outer rib width **142** of FIG. **2** is equal to the inner ring width **82**, the outer ring width **94** and/or the inner rib width **124**. The outer rib height **144** is measured radially from the support outer ring **68** at its outer ring inner side **88** to the respective outer rib distal end **130**. The outer rib height **144** may be equal to (or different than) the inner ring thickness **84**, the outer ring thickness **96** and/or the inner rib height **126**. The outer rib height **144** of FIG. **2** is less than the outer rib width **142** and the bridge height **110** of FIG. **3**. Referring to FIG. **4**, each support outer rib **74** has a lateral thickness **146** measured laterally between its opposing lateral sides **138** and **140**. The outer rib thickness **146** of FIG. **4** is greater than the inner ring thickness **84** and/or the outer ring thickness **96**. The outer rib thickness **146** may be equal to or different than the bridge thickness **112** and/or the inner rib thickness **128**. The outer rib thickness **146** of FIG. **4** is less than the outer rib width **142** of FIG. **2**. The radial gap **132** of FIG. **4** has a radial height **148** measured from the respective support inner rib **72** at its inner rib distal end **114** to the respective support outer rib **74** at its outer rib distal end **130**. The gap height **148** of FIG. **4** is less than the inner ring thickness **84**, the outer ring thickness **96**, the inner rib height **126**, the outer rib height **144**, the inner rib thickness **128** and/or the outer rib thickness **146**. The present disclosure, however, is not limited to such exemplary dimensional relationships.
(30) With the foregoing arrangement of elements **66**, **68**, **70**, **72** and **74**, the flexible support **30** is formed with a plurality of ports **150** arranged (e.g., and equispaced) circumferentially about the axis **38**. Each of these ports **150** is formed by and extends circumferentially about the axis **38** between a respective set of circumferentially aligned support ribs **72** and **74** and its respective circumferentially neighboring support bridge **70**. Each port **150** is formed by and extends radially from the support inner ring **66** at its inner ring outer side **76** to the support outer ring **68** at its outer ring inner side **88**. Referring to FIGS. **2** and **3**, each port **150** extends axially along the axis **38** through the flexible support **30** from the support first side **58** to the support second side **60**.
(31) Each port **150** of FIG. **4** has a circumferential width **152** and a radial height **154**. The port width **152** is measured circumferentially about the axis **38** from the respective set of circumferentially aligned support ribs **72** and **74** and its respective circumferentially neighboring support bridge **70**. The port width **152** of FIG. **4** is greater than the inner ring thickness **84**, the outer ring thickness **96**, the bridge thickness **112**, the bridge height **110**, the inner rib thickness **128** and/or the outer rib thickness **146**. The port height **154** is measured radially from the support inner ring **66** at its inner ring outer side **76** to the support outer ring **68** at its outer ring inner side **88**. The port height **154** of FIG. **4** may be equal to the bridge height **110** and/or a sum of the inner rib height **126**, the outer rib height **144** and the gap height **148**. The present disclosure, however, is not limited to such exemplary dimensional relationships.
(32) Referring to FIG. **1**, the flexible support **30** mounts (e.g., fixedly attaches) the bearing **28** and

its outer race **54** to the stationary structure **22** and its outer mounting land **34**. The support inner ring **66** of FIG. **1**, for example, extends axially along (e.g., axially overlaps) and circumferentially around (e.g., circumscribes) the bearing **28** and its outer race **54**. The support inner ring **66** and its interior support surface **86** further radially engage (e.g., radially contact, abut radially against, etc.) the bearing **28** and its outer race **54**. Here, the outer race **54** is attached to the support inner ring **66** by an interference fit at a radial interface between the outer race **54** and the support inner ring **66** at the support inner side **62**. Similarly, the outer mounting land **34** extends axially along and circumferentially around the flexible support **30** and its support outer ring **68**. The support outer ring **68** and its exterior support surface **98** further radially engage (e.g., radially contact, abut radially against, etc.) the stationary structure **22** and its outer mounting land **34**. Here, the support outer ring **68** is attached to the outer mounting land **34** by an interference fit at a radial interface between the support outer ring **68** at its support outer side **64** and the outer mounting land **34** and its land inner side **40**. Of course, it is contemplated the outer race **54** may be attached to the support inner ring **66** and/or the support outer ring **68** may be attached to the outer mounting land **34** using various other attachment techniques; e.g., a retention ring, one or more fasteners (e.g., bolts), etc. Moreover, it is contemplated the support inner ring **66** may be used as an outer race of the bearing **28**, and the outer race **54** may be omitted.

(33) With the foregoing arrangement, the flexible support **30** is configured to accommodate (e.g., slight) radial movement between the rotating structure **24** and the stationary structure **22** during aircraft motor operation. A portion of the support inner ring **66** between each respective circumferentially neighboring pair of the support bridges **70** (see FIG. **4**), for example, may operate as a flexible spring member. To facilitate a radial upward shift of the rotating structure **24** in FIG. **1**, each inner ring portion may deform and bow radially outward such that the respective support inner rib **72** moves radially outward towards the respective support outer rib **74**, thereby decreasing the gap height **148** of FIG. **2** or even contacting the respective support outer rib **74**. To facilitate a radial downward shift of the rotating structure **24** in FIG. **1**, each inner ring portion may deform and bow radially inwards such that the respective support inner rib **72** moves radially inwards away from the respective support outer rib **74**, thereby increasing the gap height **148** of FIG. **2**. Of course, opposite deformation of the support inner ring **66** may occur at a diametrically opposing side of the flexible support **30** to the section shown in FIG. **1**. Dimensions of the flexible support **30** may be selected to tune a spring rate of the support inner ring deformation.

(34) While the flexible support **30** is described above as being attached to the bearing **28** and the stationary structure **22** through interference fits, the present disclosure is not limited to such exemplary mounting techniques. For example, the flexible support **30** and its support outer ring **68** may also or alternatively be bonded (e.g., welded, brazed, etc.) to the stationary structure **22** and its outer mounting land **34**. In another example, referring to FIG. **5**, the flexible support **30** may also or alternatively be mechanically fastened to the stationary structure **22** and its outer mounting land **34** by one or more mechanical fasteners **156**; e.g., bolts. A mounting flange **158**, for example, may be connected to and project radially out from the support outer ring **68** at the support first side **58**/the outer ring first side **90**. This mounting flange **158** may abut axially against or otherwise axially engage the stationary structure **22** and its outer mounting land **34**. The mounting flange **158** is attached to the stationary structure **22** and its outer mounting land **34** by the mechanical fasteners **156**.

(35) In some embodiments, referring to FIG. **4**, the flexible support **30** may be configured with the circumferentially aligned support inner ribs **72** and support outer ribs **74**. In other embodiments, referring to FIG. **6**, one or more of the support outer ribs **74** (see FIG. **4**) may be omitted. The flexible support **30** of FIG. **6**, for example, is configured without any support outer ribs. Here, the radial gap **132** is formed by and extends radially between the respective support inner rib **72** at its inner rib distal end **114** and the support outer ring **68** at its outer ring inner side **88**. In still other embodiments, referring to FIG. **7**, one or more of the support inner ribs **72** (see FIG. **4**) may be

omitted. The flexible support **30** of FIG. **7**, for example, is configured without any support inner ribs. Here, the radial gap **132** is formed by and extends radially between the respective support outer rib **74** at its outer rib distal end **130** and the support inner ring **66** at its inner ring outer side **76**. Of course, in still other embodiments, it is contemplated the flexible support **30** may include a pattern of the support inner ribs **72** and the support outer ribs **74** where the inner and outer ribs **72** and **74** are circumferentially offset from one another.

(36) In some embodiments, referring to FIG. **4**, the flexible support **30** may include a single array of the support bridges **70**, a single array of the support inner ribs **72** and/or a single set of the support outer ribs **74**. In other embodiments, referring to FIG. **8**, the flexible support **30** may include multiple arrays of the support bridges **70**A and **70**B (generally referred to as "**70**"), multiple arrays of the support inner ribs **72**A and **72**B (generally referred to as "**72**") and/or multiple arrays of the support outer ribs **74**A and **74**B (generally referred to as "**74**"). The flexible support **30** of FIG. **8**, for example, includes an intermediate ring **160** radially between the support inner ring **66** and the support outer ring **68**. The support bridges **70**A, the support inner ribs **72**A and the support outer ribs **74**A are arranged between the support inner ring **66** and the support intermediate ring **160**. The support bridges **70**B, the support inner ribs **72**B and the support outer ribs **74**B are arranged between the support outer ring **68** and the support intermediate ring **160**. Each of the support bridges **70**, the support inner ribs **72** and the support outer ribs **74** may be arranged in a similar manner as described above with respect to FIG. **4**. Here, the inner support bridges **70**A of FIG. **8** are circumferentially offset from the outer support bridges **70**B. The inner support bridges **70**A of FIG. **8** are also circumferentially offset from the sets of support ribs **72**B and **74**B. Note, while each layer of the multi-layer flexible support **30** of FIG. **8** is shown with circumferentially aligned support inner ribs **72** and support outer ribs **74**, it is contemplated the inner layer and/or the outer layer may alternatively be configured with only the support inner ribs **72**A, **72**B or the support outer ribs **74**A, **74**B.

(37) In some embodiments, referring to FIG. **8**, the multi-layer flexible support **30** may be configured as a single unitary body. For example, the multi-layer flexible support **30** of FIG. **8** (like the flexible support **30** of FIG. **4**) may be cast, machined, additively manufactured and/or otherwise formed from a single mass of material as a monolithic body. In other embodiments, referring to FIG. **9**, the multi-layer flexible support **30** may be configured from multiple (e.g., coaxial and layered) segments **162** and **164**. Two separate flexible supports **30**A and **30**B, for example, may be arranged together to form the single multi-layer flexible support **30**, where each flexible support **30**A, **30**B may have a configuration similar to that described above with respect to the flexible support **30** in FIGS. **2-4** for example. The present disclosure, however, is not limited to any particular flexible support constructions or manufacturing techniques.

(38) While the flexible support **30** is described above with one or two sets of elements **70**, **72** and **74**, the present disclosure is not limited thereto. It is contemplated, for example, the flexible support **30** may alternatively include three or even more sets of the elements **70**, **72** and **74** in order to accommodate, for example, high loads.

(39) In some embodiments, referring to FIG. **1**, the flexible support **30** may be configured as an insert radially between and mounting the bearing **28** and its outer race **54** to the stationary structure **22** and its outer mounting land **34**. The flexible support **30** of the present disclosure, however, is not limited to such an exemplary flexible support configuration. For example, referring to FIG. **10**, the flexible support **30** may alternatively be configured as a bump stop support for another flexible mount **162** for the bearing **28**; e.g., a squirrel cage bearing mount. Here, the flexible support **30** and the flexible mount **162** are independently attached to the stationary structure **22**. Moreover, the flexible support **30** is spaced radially outboard from, axially overlaps and circumscribes the flexible mount **162**. With this arrangement, the flexible support **30** provides a progressive bump stop for the flexible mount **162** under certain conditions. Note, while the arrangement of FIG. **10** is shown with the single layer flexible support **30** for case of illustration, it is contemplated the arrangement of

FIG. **10** may alternatively be arranged with a multi-layer flexible support.
(40) In some embodiments, referring to FIG. **4**, the flexible support **30** may include a common (e.g., an identical) number of the bridges **70** as the respective pairs of the inner ribs **72** and the outer ribs **74**. In other embodiments, a number of bridges **70** may be different (e.g., greater or less) than a number of the pairs of the inner ribs **72** and the outer ribs **74**. For example, two or more of the bridges **70** may be disposed between each circumferentially neighboring pair of the ribs **72**, **74**. In another example, two or more of the pairs of the ribs **72**, **74** may be disposed between each circumferentially neighboring pair of the bridges **70**.
(41) In some embodiments, the gap height **148** between each respective pair of the inner ribs **72** and the outer ribs **74** may be uniform; e.g., identical. It is contemplated, however, the gap height **148** between different pairs of the inner ribs **72** and the outer ribs **74** may be different to accommodate, for example, carcass bending.
(42) The aircraft motor of the present disclosure may be configured as or otherwise include a gas turbine engine. The gas turbine engine may be a geared gas turbine engine with a geartrain operatively coupling one or more shafts to one or more rotors in a fan section, a compressor section and/or any other engine section. Alternatively, the gas turbine engine may be configured without a geartrain as a direct-drive gas turbine engine. The gas turbine engine may include a single spool or multiple spools. The gas turbine engine may be configured as a turbofan engine, a turbojet engine, a turboprop engine, a turboshaft engine, a propfan engine, a pusher fan engine or any other type of gas turbine engine. The gas turbine engine may alternatively be configured as an auxiliary power unit (APU). The present disclosure therefore is not limited to any particular types or configurations of gas turbine engines. Moreover, while the aircraft motor is described above as a gas turbine engine, it is contemplated the aircraft motor may alternatively be configured as or otherwise include another type of internal combustion (IC) engine such as, but not limited to, a rotary engine (e.g., a Wankel type engine) or a reciprocating piston engine, or even an electric motor.
(43) While various embodiments of the present disclosure have been described, it will be apparent to those of ordinary skill in the art that many more embodiments and implementations are possible within the scope of the disclosure. For example, the present disclosure as described herein includes several aspects and embodiments that include particular features. Although these features may be described individually, it is within the scope of the present disclosure that some or all of these features may be combined with any one of the aspects and remain within the scope of the disclosure. Accordingly, the present disclosure is not to be restricted except in light of the attached claims and their equivalents.

Claims

1. An assembly for an aircraft motor, comprising: a bearing extending circumferentially around an axis; a stationary structure circumscribing the bearing; and a flexible support arranged radially between and radially engaging the bearing and the stationary structure, the flexible support including a first ring, a second ring, a third ring, a plurality of first bridges, a plurality of first ribs, a plurality of second bridges and a plurality of second ribs; the second ring radially between the first ring and the third ring; the plurality of first bridges arranged circumferentially about the axis, and each of the plurality of first bridges extending radially between and connected to the first ring and the second ring; the plurality of first ribs arranged circumferentially about the axis and interspersed with the plurality of first bridges, each of the plurality of first ribs projecting radially out from the first ring towards the second ring, and each of the plurality of first ribs connected to the first ring and disengaged from the second ring; the plurality of second bridges arranged circumferentially about the axis, each of the plurality of second bridges extending radially between and connected to the second ring and the third ring, and the plurality of second bridges circumferentially offset from the plurality of first bridges; and the plurality of second ribs arranged circumferentially about the

axis and interspersed with the plurality of second bridges, each of the plurality of second ribs projecting radially out from the second ring towards the third ring, and each of the plurality of second ribs connected to the second ring and disengaged from the third ring.
2. The assembly of claim 1, wherein the first ring is an inner ring; and the second ring is an outer ring which circumscribes the inner ring.
3. The assembly of claim 1, wherein the first ring circumscribes the second ring, and the second ring circumscribes the third ring.
4. The assembly of claim 1, wherein the flexible support further includes a plurality of third ribs, each of the plurality of third ribs is connected to and projects radially out from the second ring, each of the plurality of third ribs is circumferentially aligned with a respective one of the plurality of first ribs, and each of the plurality of third ribs is separated from the respective one of the plurality of first ribs by a respective radial gap.
5. The assembly of claim 4, wherein a first of the plurality of first ribs is circumferentially aligned with a first of the plurality of third ribs; and a radial height of the first of the plurality of first ribs is equal to a radial height of the first of the plurality of third ribs.
6. The assembly of claim 4, wherein a first of the plurality of first ribs is circumferentially aligned with a first of the plurality of third ribs; and a radial height of the respective radial gap radially between and formed by the first of the plurality of first ribs and the first of the plurality of third ribs is less than at least one of a radial height of the first of the plurality of first ribs or a radial height of the first of the plurality of third ribs.
7. The assembly of claim 4, wherein a first of the plurality of first ribs is circumferentially aligned with a first of the plurality of third ribs; and a radial height of the respective radial gap radially between and formed by the first of the plurality of first ribs and the first of the plurality of third ribs is less than at least one of a lateral thickness of the first of the plurality of first ribs or a lateral thickness of the first of the plurality of third ribs.
8. The assembly of claim 1, wherein each of the plurality of first ribs is separated from the second ring by a respective radial gap.
9. The assembly of claim 8, wherein a radial height of the respective radial gap radially between and formed by a first of the plurality of first ribs and the second ring is less than at least one of a radial height of the first of the plurality of first ribs; or a lateral thickness of the first of the plurality of first ribs.
10. The assembly of claim 1, wherein a radial height of a first of the plurality of first bridges is at least one of equal to or less than a lateral width of the first of the plurality of first bridges; or greater than an axial width of the first of the plurality of first bridges.
11. The assembly of claim 1, wherein a radial height of a first of the plurality of first bridges is greater than at least one of a radial thickness of the first ring or a radial thickness of the second ring.
12. The assembly of claim 1, wherein the flexible support extends axially between a support first side and a support second side, and at least one of the first ring, the second ring, each of the plurality of first bridges and each of the plurality of first ribs projects axially to the support first side; or the first ring, the second ring, each of the plurality of first bridges and each of the plurality of first ribs projects axially to the support second side.
13. The assembly of claim 1, wherein the bearing comprises a rolling element bearing with an outer race; and one of the first ring or the third ring circumscribes and radially engages the outer race.
14. The assembly of claim 1, wherein the flexible support is attached to the stationary structure through an interference fit at a radial interface between the stationary structure and one of the first ring or the third ring.
15. The assembly of claim 1, wherein the flexible support is attached to the stationary structure through a bonded connection between the stationary structure and one of the first ring or the third ring.

16. The assembly of claim 1, wherein the flexible support is attached to the stationary structure with one or more fasteners.

17. An assembly for an aircraft motor, comprising: a bearing extending circumferentially around an axis; a stationary structure circumscribing the bearing; and a flexible support arranged radially between and radially engaging the bearing and the stationary structure, the flexible support including a first ring, a second ring, a third ring, a plurality of first bridges, a plurality of first ribs, a plurality of second bridges and a plurality of second ribs; the second ring radially between the first ring and the third ring; the plurality of first bridges arranged circumferentially about the axis, and each of the plurality of first bridges extending radially between and connected to the first ring and the second ring; the plurality of first ribs arranged circumferentially about the axis and interspersed with the plurality of first bridges, each of the plurality of first ribs projecting radially out from the second ring towards the first ring, and each of the plurality of first ribs connected to the second ring and disengaged from the first ring; the plurality of second bridges arranged circumferentially about the axis, each of the plurality of second bridges extending radially between and connected to the second ring and the third ring, and the plurality of second bridges circumferentially offset from the plurality of first bridges; and the plurality of second ribs arranged circumferentially about the axis and interspersed with the plurality of second bridges, wherein one of each of the plurality of second ribs projects radially out from the second ring towards the third ring, and each of the plurality of second ribs is connected to the second ring and is disengaged from the third ring; or each of the plurality of second ribs projects radially out from the third ring towards the second ring, and each of the plurality of second ribs is connected to the third ring and is disengaged from the second ring.